

# Exploring Wave Properties on a String

PhET Wave on a String · PhET

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|--------------------------------|-------------|-------------------|
| Senior Secondary (Years 11-12) | ~30 minutes | Science · Physics |
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**NOTICE — AI-generated worksheet**

This worksheet was drafted automatically from the site’s curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/wave-on-a-string>  
Curriculum links: waves · energy transfer · AC9S9U04

## Learning intentions

- I can identify and measure the amplitude, wavelength and frequency of a wave.
- I can explain how a wave transfers energy without net movement of the medium itself.
- I can compare wave reflection at a fixed end versus a loose (free) end.

## Getting started

Open PhET Wave on a String. Set the simulation to oscillate mode and generate a continuous wave along the string. Note the amplitude, frequency and tension controls before you start recording measurements.

## Tasks

1. Generate a continuous wave with a low frequency. Measure (using the ruler tool, or by counting) the wavelength of the wave.

2. Increase the frequency to roughly double your original value, keeping amplitude constant. Measure the new wavelength and calculate the ratio between your two wavelengths.

Working:

3. Test three different frequency settings and record the resulting wavelength for each, keeping amplitude and tension constant.

| Frequency | Wavelength |
|-----------|------------|
|           |            |
|           |            |
|           |            |

**4. Increase the amplitude while keeping frequency constant. Does this change the wavelength? Describe what does change.**

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**5. Explain, using this simulation as evidence, why a wave can transfer energy along the string without any single particle of the string travelling from one end to the other.**

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**6. Set the end of the string to 'fixed'. Send a single pulse down the string and describe what happens to the pulse when it reflects off the fixed end.**

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**7. Change the end condition to 'loose' (free end) and send the same pulse. Compare the reflected pulse to the one you saw with a fixed end, and explain why the two reflections are different.**

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***Extension (optional)***

The tension in this simulation is similar to the tension in a real guitar or violin string. Use what you've learned about frequency and wavelength to explain why tightening a string (increasing tension) raises the pitch of the note it produces.

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